

DAEN/ISEN 427 In class assignment 3

Due: Tue Oct 7 10:50 AM

M&Ms – Instructions (one copy per table)

We will use a few bags of M&Ms to estimate the unknown proportion of a particular color in a bag. Using Bayesian inference with a Beta prior and Binomial likelihood, you will update your beliefs as more data is collected and visualize how the posterior distribution changes over time.

Part 1: Prior

- Choose **one color**. Define: θ = the proportion of that color in a bag.
- Assume a Beta prior for θ : $\theta \sim \text{Beta}(\alpha_0, \beta_0)$.
Pick a prior, for instance: Uniform: $\text{Beta}(1, 1)$, Skeptical: $\text{Beta}(2, 8)$, Optimistic: $\text{Beta}(8, 2)$.
- Discuss your choice of prior and what it “encodes” about your beliefs.

Part 2: Data

- Your bag is your first sample.
- Count the total number of M&Ms (n_1) and how many are of your chosen color (x_1).
Please also count the other colors to share with your colleagues.
- Record n_1 and your x_1 values.
- For $n_{2,3,4}$ and $x_{2,3,4}$, share information with the people at your table.

Part 3: Posterior (after first sample)

Using Beta-Binomial conjugacy: $\theta \mid x_1 \sim \text{Beta}(\alpha_0 + x_1, \beta_0 + n_1 - x_1)$.

- Plot (or describe) the shape of the posterior.
- Compute posterior mean:

$$\mathbb{E}[\theta \mid \text{data}] = \frac{\alpha_0 + x_1}{\alpha_0 + \beta_0 + n_1}.$$

- Compute posterior variance/standard deviation:

$$\text{Var}(\theta \mid \text{data}) = \frac{(\alpha_0 + x_1)(\beta_0 + n_1 - x_1)}{(\alpha_0 + \beta_0 + n_1)^2(\alpha_0 + \beta_0 + n_1 + 1)}, \quad \text{sd}(\theta \mid \text{data}) = \sqrt{\frac{(\alpha_0 + x_1)(\beta_0 + n_1 - x_1)}{(\alpha_0 + \beta_0 + n_1)^2(\alpha_0 + \beta_0 + n_1 + 1)}}.$$

- Compute a 95% credible interval (CI) for θ .

If the posterior is approximately $\mathcal{N}(\mu, \sigma^2)$, then the 95% Credible Interval = $[\mu - 1.96\sigma, \mu + 1.96\sigma]$.

If posterior is sampled ... suppose you have N posterior samples $\{\theta_1, \theta_2, \dots, \theta_N\}$. A equal-tailed interval can be computed by taking the appropriate quantiles of the sampled distribution.

using python/google colab

```
import arviz as az
import numpy as np
from scipy.stats import beta

alpha_par = 9
beta_par = 3

samples = beta.rvs(alpha_par, beta_par, size=1000)
hpd_interval = az.hdi(samples, hdi_prob=0.95)
print(hpd_interval)

import matplotlib.pyplot as plt
plt.figure(figsize=(8, 4))
plt.hist(samples, bins=60, density=True, alpha=0.6, color="skyblue")
plt.axvline(lower, color="red", linestyle="--")
plt.axvline(upper, color="red", linestyle="--")
```

Reference: `scipy.stats.beta`, `arviz.hdi`.

Part 4: Posterior Update

Repeat part 3, updating n and x for two more rounds:

$$\alpha_{t+1} = \alpha_t + x_{t+1}, \quad \beta_{t+1} = \beta_t + n_{t+1} - x_{t+1}.$$

For each round:

- Update your posterior parameters.
- Recompute posterior mean and credible intervals.
- Plot the posterior after each update.

Part 5: Reflection

Write (on a Canvas discussion or your working sheet) a short paragraph addressing:

- Verify whether you can accurately predict how many M&Ms of your chosen color will be in a new bag.
- How did your posterior belief about θ evolve with more data?
- How sensitive were your results to the prior?
- Would a larger sample change your conclusions more than the prior did?
- What assumptions are implicit in using a binomial model here?